Abdallah ElTohamy

Software Engineer — Applied ML, LLM Pipelines & Speech/Audio · Native Arabic Speaker

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SUMMARY

Software engineer with 4+ years of experience shipping production systems that apply machine learning and large language models to real user problems. Built and iterated on LLM-powered transcription and audio-enhancement pipelines with AI research teams, and engineered scalable Python/Node.js services on AWS serving 5,000+ users at 99.99% uptime. Native Arabic speaker raised in Egypt with deep familiarity with the culture, dialects and idiom of the Arabic-speaking world — motivated to make Siri feel genuinely native to Arabic users.

PROFESSIONAL EXPERIENCE

Software Engineer **Oct 2023 – Present**

Shanda Studio (remote) *Washington, D.C., USA*

- **LLM & speech pipelines:** Partnered with internal AI research teams to iterate on LLM pipeline features for speech transcription and audio enhancement, feeding user-driven evaluation back into the models — cut podcast editing time by 70% and improved model performance across 1,000+ minutes of monthly audio.
- **Production ML services:** Owned customer-facing features end-to-end, designing and deploying scalable RESTful APIs in Python on AWS that serve 5,000+ active users with 99.99% uptime and latency-focused optimization.
- **Audio/media processing:** Engineered FFmpeg-based media processing, reducing file sizes by 15% and processing time by 20%, with automated metadata extraction and enrichment.
- **Data at scale:** Designed relational data models (MySQL) for authentication, subscriptions and Stripe payments, reliably managing 1,000+ user records and 200+ monthly transactions.
- **Global delivery:** Built RSS podcast hosting and syndication to Spotify and Apple Podcasts for 30+ active shows; built full-stack apps in Next.js/TypeScript and Node.js (Express).
- Delivered 100% of milestones on time as an independent remote engineer, earning a 3× scope expansion.

Software Developer (Freelance) **Jul 2020 – May 2025**

Upwork & Fiverr — Top Rated, 100% Job Success, 30 projects delivered *Cairo, Egypt*

- **Applied AI & LLM solutions:** Built AI-driven applications using TensorFlow, OpenCV and LLM pipelines for face recognition, image manipulation and automated content tagging, increasing automation efficiency by 60%.
- **Large-scale media data processing:** Created encoding and editing tooling in Python and JavaScript with FFmpeg that processed 10,000+ media files, automating 80% of manual work and improving throughput 5×.
- **Real-time systems:** Built low-latency WebSocket/Express services sustaining <100 ms response times under high traffic, and a real-time usage and performance monitor (React, Node.js, Socket.IO) that cut issue response time by 60%.
- **Data tooling:** Designed and optimized MySQL databases and Next.js admin dashboards, cutting manual data tasks by 70%.

SELECTED PROJECTS

Web-Based AI Video Editor — Multi-layer timeline editor with real-time preview and AI-powered tools (Imagen on GCP); cloud-accelerated serverless rendering reduced editing time by 50%. *React, WebGL, FFmpeg, Node.js, GCP*

Complete Greet — Real-time smart customer-interaction widget; analyzed 10K+ sessions to drive product decisions, lifting engagement 40% and satisfaction 30%. *WebSockets, Node.js, Sequelize*

Space Truss Solver — Numerical finite-element solver with real-time 3D visualization, automating structural analysis and raising productivity 90%+. *FEA, Three.js, JavaScript*

EDUCATION

Bachelor of Engineering, Cairo University — Cairo, Egypt 2024

SKILLS

Languages (spoken)	Arabic (native — Egyptian & Modern Standard Arabic), English (professional)
ML / AI	LLM pipelines, prompt & context engineering, MCP / tool-calling integrations, speech transcription & audio enhancement, TensorFlow, OpenCV, data analytics
Programming	Python, C / C++ / C#, TypeScript, JavaScript, SQL, Bash
Engineering	Scalable REST APIs, real-time systems, testing & CI/CD, Git, Docker, Linux, Nginx, FFmpeg
Cloud & Data	AWS, GCP, Azure, serverless, MySQL, Sequelize, Node.js, Next.js, React